# 05_Class_Activity

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## In-Class Activity 5: Probability and Statistical Inference

### What did we do last time?

In our previous activity, we:

- Created and interpreted frequency distributions (histograms)
- Compared data between groups using side-by-side histograms
- Explored how sample size affects our understanding of populations
- Created density plots and calculated probabilities

### Today's focus:

Today we'll focus on:

- t-distribution and when to use it
- Calculating and interpreting standard error
- Creating confidence intervals
- Conducting **one-sample and two-sample t-tests**
- **Understanding statistical assumptions and their importance**

## Setup

First, let's load the packages and data we'll be using:

```r
# Load required packages
library(readxl)
library(tidyverse)
library(patchwork)
library(car)  # For diagnostic tests

pine_df <- read_excel("data/class_pine needle length.xlsx")
pine_switch_df <- read_excel("data/class_pine needle length switched.xlsx")

head(pine_df)
```

```
# A tibble: 6 × 5
  group tree_no tree_char side  length_mm
  <chr>   <dbl> <chr>     <chr>     <dbl>
1 five        1 tree_1    sunny      22.7
2 five        1 tree_1    sunny      21.1
3 five        1 tree_1    sunny      18.6
4 five        1 tree_1    sunny      18.6
5 five        1 tree_1    sunny      21.0
6 five        1 tree_1    sunny      18.9
```

## An issue we have is pseudoreplication

What is our sample unit? branches? needles? trees? sides?

- We need to get the averages of the data so it is no psudoreplicated!
- We will group data by group, tree, tree_char, side and take the averages and save the average as length_mm
- We will also need to make dataframes of sunny and shady side of the data

```
# Now we need to average the sunny and shady sides as these are all pseudoreplicated
p_df <- pine_df %>%
  group_by(group, tree_no, tree_char, side) %>%
  summarise(length_mm = mean(length_mm, na.rm=TRUE))
```

```
`summarise()` has grouped output by 'group', 'tree_no', 'tree_char'. You can
override using the `.groups` argument.
```

```
ps_df <- pine_switch_df %>%
  group_by(group, tree_no, tree_char, side) %>%
  summarise(length_mm = mean(length_mm, na.rm=TRUE))
```

```
`summarise()` has grouped output by 'group', 'tree_no', 'tree_char'. You can
override using the `.groups` argument.
```

```
ps_shady_df <- ps_df %>%
  filter(side == "shady")

ps_sunny_df <- ps_df %>%
  filter(side == "sunny")

head(ps_shady_df) %>% arrange(tree_no)
```

```
# A tibble: 6 × 5
# Groups:   group, tree_no, tree_char [6]
  group                       tree_no tree_char side  length_mm
  <chr>                         <dbl> <chr>     <chr>     <dbl>
1 five                              1 tree_1    shady      20.3
2 big_fat_fecund_female_fish        2 tree_2    shady      15.4
3 bill                              3 tree_3    shady      16.7
4 ciabatta                          5 tree_5    shady      19.1
5 moose_walkin                      7 tree_7    shady      20.7
6 fake_data                         8 tree_8    shady      17.4
```

## Part 1: Exploring the Data

Before conducting statistical tests, it’s important to understand your data.

Practice Exercise 1: Summary data

Let's create summary data of needle lengths from each side to visualize their distributions.

```
stats_df <- ps_df %>%
  group_by(side) %>%
  summarize(
    mean_length = mean(length_mm, na.rm = TRUE),
    sd_length = sd(length_mm, na.rm = TRUE),
    se_length = sd(length_mm, na.rm = TRUE)/ sum(!is.na(length_mm))^.5,
    count = sum(!is.na(length_mm)),
    .groups = "drop"
  )
stats_df
```

```
# A tibble: 2 × 5
  side  mean_length sd_length se_length count
  <chr>       <dbl>     <dbl>     <dbl> <int>
1 shady        17.6      2.51     0.886     8
2 sunny        16.2      2.64     0.934     8
```

```
# CAN YOU THINK OF AN EASIER WAY?
```

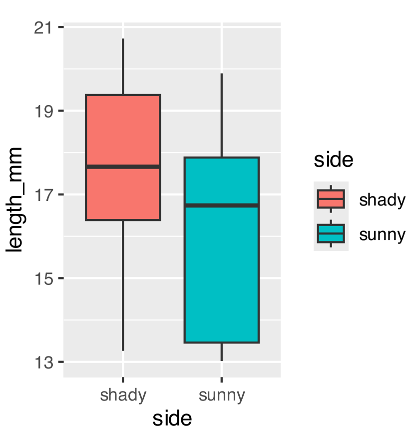
## Part 1: Exploring the Data

Before conducting statistical tests, it's important to understand your data.

Practice Exercise 1: Creating Histograms

Let's create histograms of needle lengths from each lake to visualize their distributions.

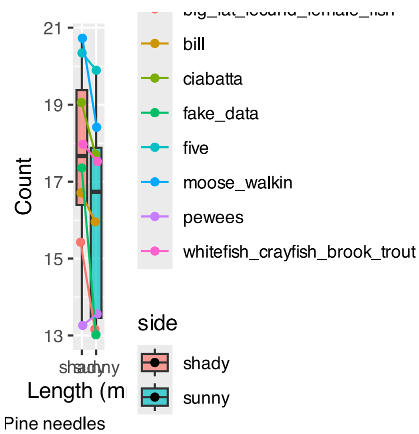
```
needle_box_plot <- ps_df %>%
ggplot(aes(x=side, y=length_mm, fill = side))+
geom_boxplot()
needle_box_plot
```



## Part 2: Exploring data a different way

We want to see variation in a different way and how groups measured data on sunny and shady sides

```r
pd <- position_dodge2(width = 0.2)
p_plot <- ps_df %>%
  ggplot(aes(side, length_mm, fill=side)) +
  geom_boxplot(alpha = 0.7) +
  geom_point(aes(group = group, color= group),
             position = pd)+
  geom_line(aes(group = group, color= group),
             position = pd)+
  labs(x = "Length (mm)", y = "Count", caption = "Pine needles")
p_plot
```




## Part 3: Testing Assumptions

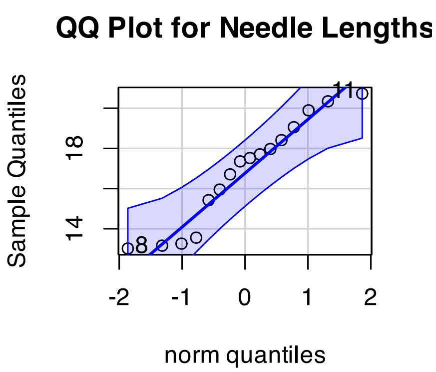
Before conducting a t-test, we need to check if our data meets the necessary assumptions:

1. **Normality**: The data should be approximately normally distributed
2. **Independence**: Observations should be independent
3. **No extreme outliers**: Outliers can heavily influence t-test results

Let's check the normality assumption for shady needle lengths:

Practice Exercise 2: Checking Normality of all data but does not do by group!

```
# Create a QQ plot to check normality
# QQ plots compare our data to a theoretical normal distribution
# Points should roughly follow the line if data is normally distributed
qqPlot(ps_df$length_mm,
       main = "QQ Plot for Needle Lengths",
       ylab = "Sample Quantiles")
```




```
[1]  8 11
```

## Shapiro-Wilk test on all data

What do we look at here and how do we interpret?

Hoping for non significant!!!

```
# Also perform a formal test of normality using the Shapiro-Wilk test
# Null hypothesis: Data is normally distributed
# If p > 0.05, we don't reject the assumption of normality
shapiro_test <- shapiro.test(ps_df$length_mm)
print(shapiro_test)
```

```

	Shapiro-Wilk normality test

data:  ps_df$length_mm
W = 0.92754, p-value = 0.2228
```

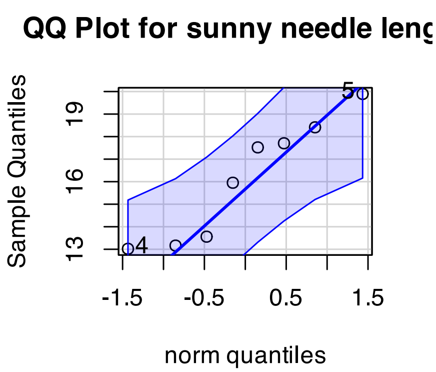
## QQ Plot for Sunny Data

Practice Exercise 8: Test normality of pine needle lengths on sunny side

qqplots

Note you need to test each groups separately...

```
# QQ Plot for sunny group
qqPlot(ps_sunny_df$length_mm,
       main = "QQ Plot for sunny needle length",
       ylab = "Sample Quantiles")
```




```
[1] 5 4
```

## Shapiro-Wilk Test for Sunny data

Practice Exercise 9: Test normality of sunny needle lengths

Shapiro-Wilk test

Note you need to test each groups separately...

```
# Shapiro-Wilk test for sunny group
shapiro.test(ps_sunny_df$length_mm)
```

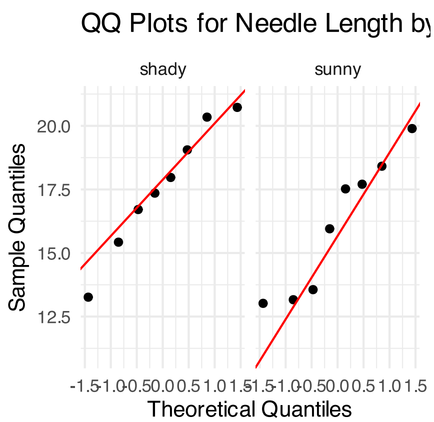
```

	Shapiro-Wilk normality test

data:  ps_sunny_df$length_mm
W = 0.89994, p-value = 0.2886
```

## Combined QQ plots with GGPLOT

We can use ggplot which you are more familiar with to also do qqPlots

```
ggplot(ps_df, aes(sample = length_mm)) +
  stat_qq() +
```

```r
  stat_qq_line(color = "red") +
  facet_wrap(~ side) +
  labs(title = "QQ Plots for Needle Length by Side",
       x = "Theoretical Quantiles",
       y = "Sample Quantiles") +
  theme_minimal()
```




## Combined Normality Test

Note you can do this a lot easier with piping of the data...

> Practice Exercise 12: Test Normality at one time
>
> There are always a lot of ways to do everything in R and is sometimes frustrating

```r
# there are always two ways
# Test for normality using Shapiro-Wilk test for each wind group
# All in one pipeline using tidyverse approach
normality_results <- ps_df %>%
  group_by(side) %>%
  summarize(
    shapiro_stat = shapiro.test(length_mm)$statistic,
    shapiro_p_value = shapiro.test(length_mm)$p.value,
    normal_distribution = if_else(shapiro_p_value > 0.05, "Normal", "Non-normal")
    # above wwe are using an ifelse test which is a great oneliner
  )

# Print the results
print(normality_results)
```

```
# A tibble: 2 × 4
  side  shapiro_stat shapiro_p_value normal_distribution
  <chr>        <dbl>           <dbl> <chr>
1 shady        0.966           0.868 Normal
2 sunny        0.900           0.289 Normal
```

## Test for Equal Variances of Sunny and Shady

Practice Exercise 13: Test equal variances

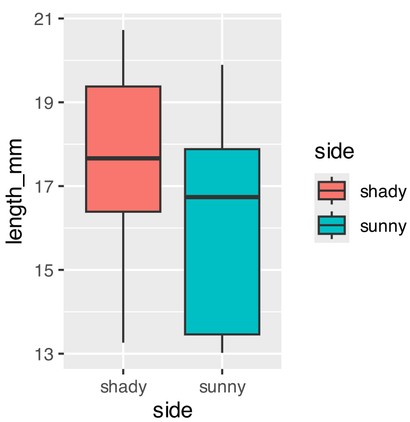
Levene's test can be done on the original dataframe

```
# Method 1: Using car package's leveneTest
# This is often preferred as it's more robust to departures from normality
levene_result <- leveneTest(length_mm ~ side, data = ps_df)
levene_result
```

```
Levene's Test for Homogeneity of Variance (center = median)
      Df F value Pr(>F)
group  1  0.2062 0.6567
      14
```

## Check for outliers

```
# Check for outliers using a boxplot
needle_box_plot
```




Tip

How to interpret these results:

- The QQ plot: Points should follow the straight line if data is normally distributed
- Shapiro-Wilk test: If p > 0.05, we don't reject the assumption of normality
- Boxplot: Look for points beyond the whiskers as potential outliers

## Part 4: One-Sample t-Test

A one-sample t-test compares a sample mean to a specific value.

Let's test if the mean needle length on the shady side is 15 or not?

Practice Exercise: One-Sample t-Test

```r
# Calculate the mean of shady needles?
shady_mean <- mean(ps_shady_df$length_mm, na.rm=TRUE)
shady_mean
```

```
[1] 17.60561
```

```r
# what is the mean
# 17.60561

ps_shade_mean <- mean(ps_shady_df$length_mm, na.rm = TRUE)
cat("Mean:", round(ps_shade_mean, 1), "mm\n")
```

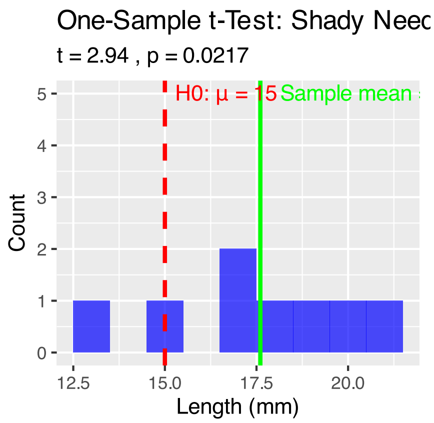
```
Mean: 17.6 mm
```

```r
# Perform a one-sample t-test
t_test_result <- t.test(ps_shady_df$length_mm, mu = 15)

# View the test results
t_test_result
```

```

	One Sample t-test

data:  ps_shady_df$length_mm
t = 2.9414, df = 7, p-value = 0.02167
alternative hypothesis: true mean is not equal to 15
95 percent confidence interval:
 15.51092 19.70030
sample estimates:
mean of x
 17.60561
```

```r
# Create a visualization of the test
ps_shady_df %>%
  ggplot(aes(x = length_mm)) +
  geom_histogram(binwidth = 1, fill = "blue", alpha = 0.7) +
  geom_vline(xintercept = 15, color = "red", linetype = "dashed", linewidth = 1) +
  geom_vline(xintercept = shady_mean, color = "green", linewidth = 1) +
  annotate("text", x = 15, y = 5, label = "H0: μ = 15", color = "red", hjust = -0.1) +
  annotate("text", x = ps_shade_mean, y = 5,
           label = paste("Sample mean =", round(ps_shade_mean, 1)),
           color = "green", hjust = -0.1) +
  labs(
    title = "One-Sample t-Test: Shady Needle Lengths",
    subtitle = paste(
      "t =", round(t_test_result$statistic, 2),
      ", p =", format.pval(t_test_result$p.value, digits = 3)),
```

```
    x = "Length (mm)",
    y = "Count")
```




Tip

Interpret the results:

1. What was the null hypothesis? H0: μ = 15mm
2. What was the alternative hypothesis? H1: μ ≠ 15mm
3. What does the p-value tell us? (Is $p < 0.05$?)
4. Should we reject or fail to reject the null hypothesis?
5. What is the practical interpretation for biologists?

## Part 5: Confidence Intervals

A confidence interval gives us a range of plausible values for the population mean.

For a 95% confidence interval using the t-distribution:

$$95\% \quad \text{CI} = \bar{x} \pm t_{\alpha/2,n-1} \times \frac{s}{\sqrt{n}}$$

Where:

- $\bar{x}$ is the sample mean
- $s$ is the sample standard deviation
- $n$ is the sample size
- $t_{\alpha/2,n-1}$ is the critical t-value with n-1 degrees of freedom

**Practice Exercise 4: Calculating Confidence Intervals**

Let's calculate the 95% confidence interval for shady needle lengths:

```
# Extract sample statistics
shady_stats <- ps_shady_df
shady_mean <- mean(ps_shady_df$length_mm, na.rm=TRUE)
shady_se <- sd(ps_shady_df$length_mm, na.rm=TRUE)/sum(!is.na(ps_shady_df$length_mm))^.5
shady_n <- sum(!is.na(ps_shady_df$length_mm))

# Find the critical t-value for 95% confidence with n-1 degrees of freedom
# qt(0.975, df) gives the t-value for a 95% confidence interval (two-tailed)
t_critical <- qt(0.975, df = shady_n - 1)
cat("Critical t-value for", shady_n-1, "degrees of freedom:", round(t_critical, 3),
"\n")
```

```
Critical t-value for 7 degrees of freedom: 2.365
```

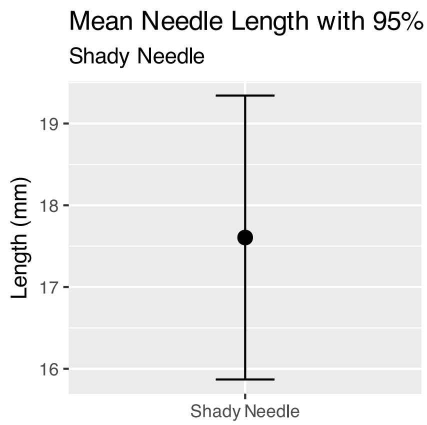
```
# Calculate the confidence interval
shady_ci_lower <- shady_mean - t_critical * shady_se
shady_ci_upper <- shady_mean + t_critical * shady_se

# Display the confidence interval
cat("95% Confidence Interval for Shady needle mean length:",
    round(shady_ci_lower, 1), "to", round(shady_ci_upper, 1), "mm\n")
```

```
95% Confidence Interval for Shady needle mean length: 15.5 to 19.7 mm
```

```
# Compare this to a confidence interval using the normal approximation (z = 1.96)
z_ci_lower <- shady_mean - 1.96 * shady_se
z_ci_upper <- shady_mean + 1.96 * shady_se

cat("95% CI using normal approximation:",
    round(z_ci_lower, 1), "to", round(z_ci_upper, 1), "mm\n")
```

```
95% CI using normal approximation: 15.9 to 19.3 mm
```

```
# Visualize the confidence interval
ggplot() +
  geom_errorbar(aes(x = "Shady Needle",
                    ymin = z_ci_lower,
                    ymax = z_ci_upper),
                width = 0.2) +
  geom_point(aes(x = "Shady Needle", y = shady_mean), size = 3) +
  labs(title = "Mean Needle Length with 95% Confidence Interval",
       subtitle = "Shady Needle",
       x = NULL,
       y = "Length (mm)")
```




## Part 6: Two-Sample t-Test

A two-sample t-test compares means from two independent groups.

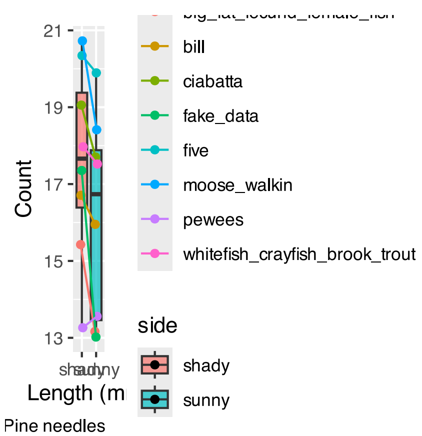
Let's compare needle lengths between sunny and shady sides

```
# Summarize pine needle data by wind exposure
stats_df
```

```
# A tibble: 2 × 5
  side  mean_length sd_length se_length count
  <chr>       <dbl>     <dbl>     <dbl> <int>
1 shady        17.6      2.51     0.886     8
2 sunny        16.2      2.64     0.934     8
```

## Look a the plot of needle lengths

```
# Create a boxplot to visualize the data
p_plot
```



## Now let's conduct the two-sample t-test:

Practice Exercise 6: Two-Sample t-Test

### Calculate the T Test

```r
# Perform a two-sample t-test
# H0: μ1 = μ2 (The mean needle lengths are equal)
# H1: μ1 ≠ μ2 (The mean needle lengths are different)

# var.equal=TRUE uses the standard t-test (pooled variance)
# var.equal=FALSE uses Welch's t-test (for unequal variances)
t_test_result <- t.test(length_mm ~ side, data = ps_df, var.equal = TRUE)

# Display the test results
t_test_result
```

```

	Two Sample t-test

data:  length_mm by side
t = 1.1279, df = 14, p-value = 0.2783
alternative hypothesis: true difference in means between group shady and group sunny is
not equal to 0
95 percent confidence interval:
 -1.309330  4.214005
sample estimates:
mean in group shady mean in group sunny
           17.60561            16.15328
```

### Compare to a Welch's Two Sample T Test when variance is unequal

```r
# Perform a two-sample t-test
# H0: μ1 = μ2 (The mean needle lengths are equal)
# H1: μ1 ≠ μ2 (The mean needle lengths are different)

# var.equal=TRUE uses the standard t-test (pooled variance)
# var.equal=FALSE uses Welch's t-test (for unequal variances)
welch_test_result <- t.test(length_mm ~ side, data = ps_df, var.equal = FALSE)

# Display the test results
welch_test_result
```

```

	Welch Two Sample t-test

data:  length_mm by side
t = 1.1279, df = 13.96, p-value = 0.2784
alternative hypothesis: true difference in means between group shady and group sunny is not
equal to 0
95 percent confidence interval:
 -1.310069  4.214743
sample estimates:
```

```
mean in group shady mean in group sunny 
           17.60561            16.15328 
```

## Compare to a Paired T Test

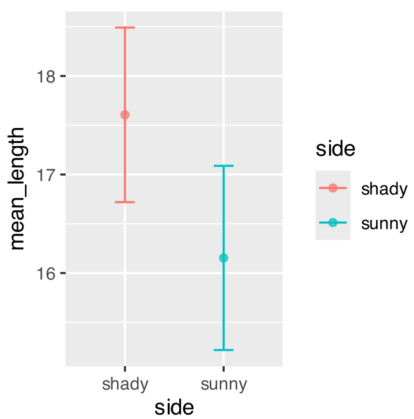
```
# Perform a two-sample t-test
# H0: µ1 = µ2 (The mean needle lengths are equal)
# H1: µ1 ≠ µ2 (The mean needle lengths are different)

# var.equal=TRUE uses the standard t-test (pooled variance)
# var.equal=FALSE uses Welch's t-test (for unequal variances)
paired_test_result <- t.test(length_mm ~ side, data = ps_df, pair = TRUE)

# Display the test results
paired_test_result
```

```

	Paired t-test

data:  length_mm by side
t = 2.7818, df = 7, p-value = 0.02723
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 0.2178092 2.6868652
sample estimates:
mean difference 
       1.452337 
```

## Visualize the plot

Tip

```
# Visualize the results with a mean and error bar plot
ggplot(stats_df, aes(x = side, y = mean_length, color=side, fill = side)) +
  geom_point(stat = "identity", alpha = 0.7) +
  geom_errorbar(aes(ymin = mean_length - se_length,
                    ymax = mean_length + se_length),
                width = 0.2)
```



### Interpretation:

> Tip
>
> Interpret the results:
>
> 1. What was the null hypothesis?
> 2. What was the alternative hypothesis?
> 3. What does the p-value tell us?
> 4. Should we reject or fail to reject the null hypothesis?
> 5. What is the practical interpretation for botanists?

## Part 8: Communicating Statistical Results

In scientific writing, it's important to report statistical results clearly and consistently.

Here's a standard format for reporting t-test results:

For a one-sample t-test: "A one-sample t-test showed that the mean needle length shady side (M = [mean], SD = [sd]) was [significantly/not significantly] different from 15 mm, t([df]) = [t-value], p = [p-value]."

For a two-sample t-test: "A two-sample t-test revealed that pine needle lengths on the sunny side (M = [mean1], SD = [sd1]) were [significantly/not significantly] [longer/shorter] than on the shady side (M = [mean2], SD = [sd2]), t([df]) = [t-value], p = [p-value]."

> Practice Exercise 8: Writing Statistical Results
>
> Write properly formatted statements reporting the results of:
>
> 1. The one-sample t-test comparing mean shady needle lenght to 15mm
> 2. The two-sample t-test comparing pine needle lengths
> 3. The two-sample t-test comparing needle lengths between sides
>
> Remember to include:
>
> - Means and standard deviations for each group
> - The t-value with degrees of freedom
> - The p-value and whether the result is significant

## Reflection Questions

1. How does the t-distribution differ from the normal distribution, and why does this matter for small samples?
2. What assumptions must be met to use a t-test, and what alternatives exist if these assumptions are violated?
3. What is the difference between statistical significance and practical importance?
4. How would the confidence interval change if we used a 99% confidence level instead of 95%?
5. How would you explain the concept of a p-value to someone with no statistical background?